

Functional Dependencies

General

Term	Description
Superkey	Any set of attributes that can uniquely identify rows. May contain extra (redundant) attributes.
Candidate Key	A minimal superkey — no attribute can be removed without losing uniqueness.

Walk-through

1. Identify Keys and Superkeys

- A **superkey** is a set of attributes that functionally determine all other attributes.
- A **candidate key** is a minimal superkey.

How?

- Find **attribute closures**: for a set X , compute X^+ (closure under FDs).
- If X^+ contains all attributes of the relation, then X is a superkey.
- Check for minimality (remove redundant attributes from X).

2. Check Normal Forms

Use the following hierarchy:

First Normal Form (1NF): No multivalued attributes (usually assumed).

Second Normal Form (2NF):

- Must be in 1NF.
- No **partial dependency** on a **candidate key** (i.e. no proper subset of a key determines a non-prime attribute).

Third Normal Form (3NF):

- Must be in 2NF.
- No **transitive dependency** on a candidate key.
- A FD $X \rightarrow A$ violates 3NF if:
 - X is **not a superkey**, and
 - A is **not** a prime attribute (i.e. not part of any candidate key).

Boyce-Codd Normal Form (BCNF):

- Stronger than 3NF.
- Every non-trivial FD $X \rightarrow A$ must have X as a **superkey**.

Steps to Normalize to 3NF/BCNF

Step 1: Find all candidate keys

Use closures of attribute combinations.

Step 2: Check BCNF violations

- For each FD $X \rightarrow Y$, check:
 - Is X a superkey?  → OK
 -  → Violation → **Decompose**

Step 3: Decomposition (if not in BCNF)

- If $X \rightarrow Y$ violates BCNF:
 - Create two relations:
 1. $R_1 = X \cup Y$
 2. $R_2 = R - (Y - X)$
- Repeat the process on each new relation.

Step 4: If only 3NF is needed (e.g. for preserving dependencies):

- Use **3NF synthesis algorithm**:
 1. Compute minimal cover of FDs.
 2. Create a relation for each FD.
 3. Ensure all candidate keys are preserved.

Tricks & Tips

Problem Type	What to Do
Given FDs, find keys	Use closures to find minimal superkeys
Given relation, normalize	Check each FD against BCNF definition
Checking "is in BCNF?"	Make sure all FD LHS are superkeys
Checking "is in 3NF?"	If not superkey, RHS must be prime
"Exactly two/three relations" in result?	Track decompositions made
Trivial FDs?	$X \rightarrow Y$ where $Y \subseteq X$ are trivial
FD: $A \rightarrow ABCDE$	A is a superkey if $A^+ = \text{all attributes}$

Example Mini-Walkthrough

Given:

$R(A, B, C, D, E)$

FDs:

- $AB \rightarrow CDE$
- $C \rightarrow AD$
- $D \rightarrow E$

1. Candidate keys?

Try AB^+ : $AB \rightarrow CDE \rightarrow A, B, C, D, E$  $\rightarrow AB$ is key

Check for minimality (A alone? A^+ ? No $\rightarrow$ So AB is minimal)

2. Check FDs for BCNF violation

- $AB \rightarrow CDE$  LHS is key $\rightarrow$ OK

- $C \rightarrow AD$ ✗ C not a key $\rightarrow$ Violation
- $D \rightarrow E$ ✗ D not a key $\rightarrow$ Violation

3. **Decompose**

From $C \rightarrow AD$ $\rightarrow$ make $R1(C, A, D)$, and rest becomes $R2(C, B, E)$

Repeat for $R2$ if needed